PROJECT DESIGN PHASE

Solution Architecture

Date	02 October 2026	**Team ID**	SWTID-2026-9980
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value		

1. Architecture Overview

The proposed system evaluates access requests against configurable field-value rules. A script-controlled ACL checks the requesting user, the requested operation, and the current record field value before granting or denying access.

Rule inputs: user permissions • target record • configured field value • requested action

2. Components and Responsibilities

Component	Responsibility
Requester / Application	Initiates a read or write request for a specific record.
Access Request	Carries the user identity, target record, requested operation, and relevant record data.
ACL Script Engine	Runs the configured script, evaluates field-value conditions, and checks the applicable permission rules.
Decision Handler	Returns an allow or deny decision. Unauthorized requests are blocked; authorized requests proceed.
Record Store / Target System	Provides the record for evaluation and applies the operation only after authorization.

3. Access-Control Flow

1) A user or application requests an operation on a record. 2) The system identifies the requester, target record, and requested action. 3) The ACL script reads the relevant field value and evaluates the configured condition together with the user’s permissions. 4) If the rule is satisfied, the operation is allowed; otherwise, the request is denied. 5) The decision can be logged for auditing and troubleshooting.

4. Security and Implementation Considerations

Apply authorization checks on every protected request; use least-privilege permissions; define clear behavior for missing or invalid field values; test both allowed and denied cases; and maintain rule documentation and change history. The architecture is platform-agnostic and should be adapted to the selected record-management platform’s ACL and scripting capabilities.